Placing Fractions on a Number Line
Answer Key

Grade 2–3   Fractions as Numbers

Quick Answers

1. $\frac{1}{2}$
2. $\frac{3}{4}$
3. $\frac{2}{3}$
4. $\frac{2}{3}$
5. 1
6. —
7. —
8. $\frac{3}{2}$
9. $\frac{2}{3}$
10. —

1. Point A on a line cut into halves.

Step 1. Count the equal parts between 0 and 1. There are two of them, so every part is one half and the bottom number is 2.
Step 2. Count the parts travelled from 0 to reach A: just one. So the top number is 1.
Step 3. One part out of two equal parts is $\boxed{\frac{1}{2}}$.
Teaching note: the tick at A is also the halfway point of the whole line, which is a good visual check before any counting starts.

2. Point B on a line cut into fourths.

Step 1. There are four equal parts between 0 and 1, so each part is one fourth and the bottom number is 4.
Step 2. From 0, the parts travelled to reach B are: one, two, three. The top number is 3.
Step 3. Three parts out of four is $\boxed{\frac{3}{4}}$.
Common wrong answer: $\frac{3}{5}$ — the five *tick marks* were counted instead of the four equal *parts*. Have the child colour in the parts, not the ticks.

3. Point C on a line cut into thirds.

Step 1. Three equal parts between 0 and 1 makes each part one third, so the bottom number is 3.
Step 2. Counting from 0: one part, two parts — that reaches C. The top number is 2.
Step 3. The point is $\boxed{\frac{2}{3}}$.
Common wrong answer: $\frac{1}{3}$ — the one part left *after* C was counted. Counting always starts at 0 and runs up to the point.

4. Placing $\frac{4}{6}$ and renaming it.

Step 1. The line is cut into six equal parts, so each tick is one sixth. To place $\frac{4}{6}$, count four parts from 0: that is the fourth tick after 0. Mark it P.
Step 2. Now look at the same picture in bigger pieces: two sixths make one third, so the six parts group into three equal thirds. Point P lands exactly on the boundary of the second group.
Step 3. Two thirds out of three, so $\frac{4}{6}$ and $\boxed{\frac{2}{3}}$ name the same point.
Listen for: "it moved" — the point does not move. Only the size of the parts we count with changes.

5. Placing $\frac{8}{8}$.

Step 1. Each part on this line is one eighth, and there are eight of them between 0 and 1.
Step 2. Counting eight parts from 0 uses up every part, which lands on the far end of the whole.
Step 3. The far end of the whole is the number $\boxed{1}$.
Common wrong answer: $\frac{7}{8}$ — a child who believes a fraction can never quite reach a whole stops one tick short. Eight eighths *is* one.

6. Comparing $\frac{2}{3}$ with $\frac{2}{4}$.

Step 1. Both fractions travel the same *number* of parts (two), so the comparison is decided entirely by how long one part is.
Step 2. Cutting one whole into three parts makes longer parts than cutting it into four. On the pictures, $\frac{2}{3}$ sits past the middle of the line and $\frac{2}{4}$ sits exactly on the middle.
Step 3. $\frac{2}{3}$ is farther from 0, so $\frac{2}{3} \boxed{>} \frac{2}{4}$.
Listen for: "four is more than three, so $\frac{2}{4}$ is bigger." The bottom number counts how many pieces the whole was broken into — more pieces means smaller pieces.

7. Comparing $\frac{5}{8}$ with $\frac{3}{4}$.

Step 1. The two fractions are written in different units, so put them on one line first. Every fourth is two eighths.
Step 2. $\frac{3}{4}$ is therefore three groups of two eighths, which is the sixth tick. $\frac{5}{8}$ is the fifth tick.
Step 3. The fifth tick comes before the sixth, so $\frac{5}{8} \boxed{<} \frac{3}{4}$.
Teaching note: rewriting one fraction so both share a unit is the whole idea behind common denominators, met here as a picture.

8. Point D on a line that runs to 2.

Step 1. Check the labels first: this line runs from 0 to 2, and each whole is cut into halves, so every part is one half.
Step 2. Count halves from 0 to D: one half, two halves (that is 1), three halves. The top number is 3 and the bottom number is 2.
Step 3. Point D is $\boxed{\frac{3}{2}}$, which is one whole and one more half.
Common wrong answer: $\frac{3}{4}$ — the line was read as if it ended at 1, so all four parts were treated as parts of a single whole. The label under the middle tick says 1, and that settles it.

9. Placing $\frac{5}{3}$ and measuring past 1.

Step 1. Each part here is one third, and 1 sits at the third tick. To place $\frac{5}{3}$, count five thirds from 0 — that is the fifth tick. Mark it E.
Step 2. The question asks how far E sits *past* 1, so count from the tick labelled 1 instead of from 0: fourth tick, fifth tick. That is two parts.
Step 3. Two parts of a whole cut into thirds is $\boxed{\frac{2}{3}}$ past 1. (Written as a subtraction: $\frac{5}{3} - 1 = \frac{2}{3}$.)
Common wrong answer: $\frac{5}{3}$ — the whole distance from 0 was given instead of the distance past 1.

10. What Ellie got wrong. *Open explanation — flagged for manual review; the model response below is what a full-credit answer contains.*

Step 1. Mark the points. On the thirds line, $\frac{1}{3}$ is the first tick after 0. On the halves line, $\frac{1}{2}$ is the first tick after 0, which is the middle of the whole.

Step 2. Compare the two distances. $\frac{1}{3}$ stops short of the middle; $\frac{1}{2}$ lands on it. So $\frac{1}{2}$ is farther from 0.

Step 3. Say why. Both fractions travel one part, but the parts are not the same size: cutting a whole into 3 pieces gives *smaller* pieces than cutting it into 2. A bigger bottom number means more pieces, and more pieces means each one is shorter — so $\frac{1}{3} < \frac{1}{2}$.

Full credit: the two points are marked in the right places *and* the sentence says that a larger denominator makes smaller parts. A sentence that only says "because I looked at the picture" has the right answer without the reason — ask the child to finish the thought.